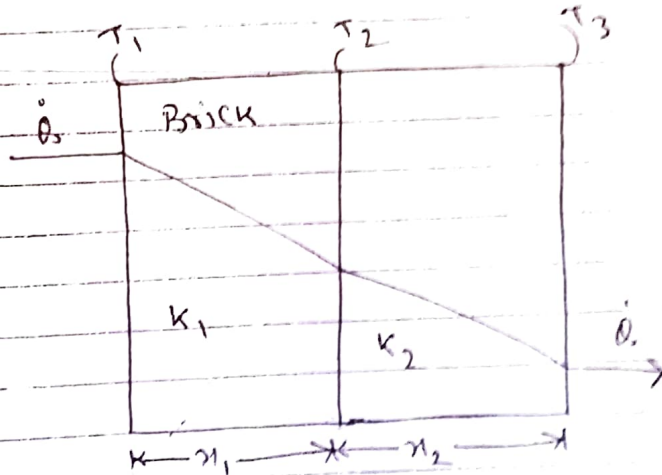


HEAT TRANSFER

1. A brick wall 450 mm thick is plastered with concrete 30 mm thick. The thermal conductivity of the brick and the concrete are 0.7 W/mK and 0.92 W/mK , respectively. If the temperature of the exposed brick face is 35°C and that of concrete is 10°C , determine the heat loss per hour through the wall which is 5 m long and 3.6 m high. Also determine the interface temperature.



$$x_1 = 0.45 \text{ m}$$

$$x_2 = 0.03 \text{ m}$$

$$k_1 = 0.7 \text{ W/mK}$$

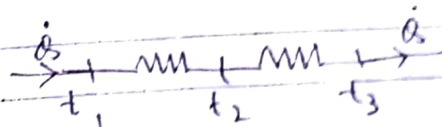
$$k_2 = 0.92 \text{ W/mK}$$

$$t_1 = 35^\circ\text{C}$$

$$t_3 = 10^\circ\text{C}$$

$$l = 5 \text{ m}$$

$$b = 3.6 \text{ m}$$



$$A = l \times b$$

$$= 5 \times 3.6$$

$$= 18 \text{ m}^2$$

$$\dot{Q} = \frac{t_1 - t_2}{\frac{x_1}{k_1 A}} = \frac{t_2 - t_3}{\frac{x_2}{k_2 A}}$$

By question,

$$\dot{Q} = \frac{t_1 - t_3}{\frac{x_1}{k_1 A} + \frac{x_2}{k_2 A}}$$

$$= \frac{35 - 10}{\frac{0.45}{0.7 \times 18} + \frac{0.03}{0.92 \times 18}}$$

$$= \frac{25}{0.0357 + 0.00181}$$

$$= \boxed{666.48} \text{ J/s}$$

$$= (666.48 \times 60 \times 60) \text{ J/h}$$

$$= \frac{666.48 \times 60 \times 60}{1000} \text{ kJ/h}$$

$$= 2399.32 \text{ kJ/h} \quad \underline{\underline{\text{Ans}}}$$

Again,

$$\dot{Q} = \frac{T_1 - T_2}{\frac{m_1}{k_1 A}}$$

$$666.48 = \frac{35 - T_2}{\frac{0.45}{0.7 \times 18}}$$

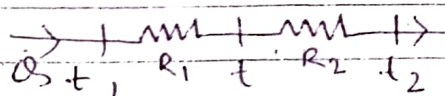
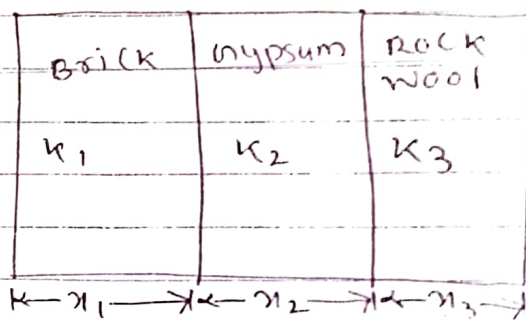
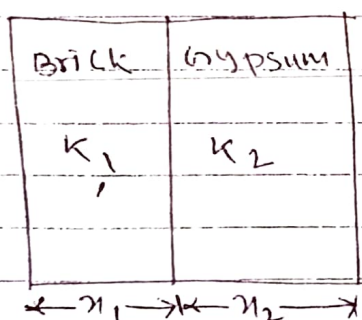
$$\S \quad 35 - T_2 = 666.48 \times 0.0357$$

$$\S \quad 35 - 23.79 = T_2$$

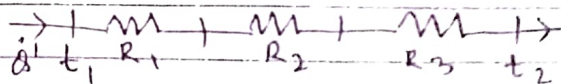
$$T_2 = 11.2^\circ \text{C} \quad \underline{\underline{\text{Ans}}}$$

Q12) An exterior wall of a house may be approximated by a 100 mm layer of common brick ($k = 0.7 \text{ W/mK}$) followed by a 4 mm layer of gypsum plaster ($k = 0.48 \text{ W/mK}$). What thickness of loosely packed rock-wool insulation ($k = 0.065 \text{ W/mK}$) should be added to reduce the heat loss (or gain) through the wall by 80%? [P.U. 2007 Fall]

Solⁿ:-



case (I)



case (II)

Case (I) :- considering no rock-wool insulation, we have,

$$\dot{Q} = \frac{t_1 - t_2}{R_1 + R_2} \quad \dots \dots (i)$$

Case (II) :- considering also the rock-wool insulation, we have,

$$\dot{Q}' = \frac{t_1 - t_2}{R_1 + R_2 + R_3} \quad \dots \dots (ii)$$

Dividing (i) by (ii) we get,

$$\frac{\dot{Q}}{\dot{Q}'} = \frac{t_1 - t_2}{R_1 + R_2} \times \frac{R_1 + R_2 + R_3}{t_1 - t_2} = \frac{R_1 + R_2 + R_3}{R_1 + R_2} \quad \text{--- (III)}$$

By question we have,

$$\begin{aligned}\dot{Q}' &= \dot{Q} - 8005 \dot{Q} \\ &= \dot{Q} - 0.8 \dot{Q} \\ &= 0.2 \dot{Q}\end{aligned}$$

$$\therefore \frac{\dot{Q}}{\dot{Q}'} = \frac{1}{0.2} = 5 \quad \text{--- (iv)}$$

From (III) & (iv) we have,

$$\frac{R_1 + R_2 + R_3}{R_1 + R_2} = 5$$

$$\S R_1 + R_2 + R_3 = 5R_1 + 5R_2$$

$$\S R_3 = 4(R_1 + R_2)$$

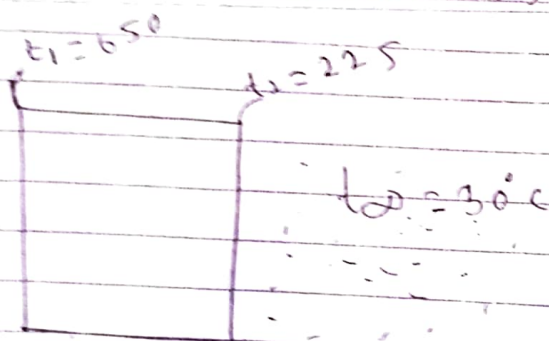
$$\S \frac{\eta_3}{k_3 A} = 4 \left(\frac{\eta_1}{k_1 A} + \frac{\eta_2}{k_2 A} \right) \quad \times \frac{1}{A}$$

$$\begin{aligned}\eta_3 &= 4k_3 \left(\frac{\eta_1}{k_1} + \frac{\eta_2}{k_2} \right) \\ &= 4 \times 0.065 \left(\frac{100}{0.7} + \frac{40}{0.48} \right) \\ &= 0.26 (226.190)\end{aligned}$$

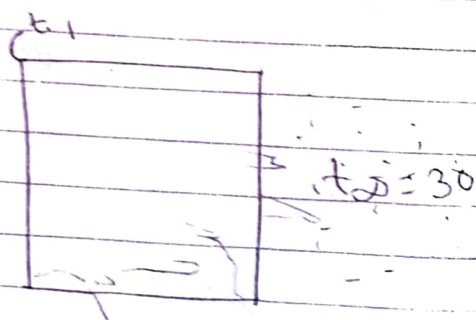
$$\therefore \eta_3 = 58.80 \text{ mm } \underline{\text{Ans}}$$

(3) The temperature at the inside and outside surfaces of the brick-work of furnace have been noted to be 650°C and 225°C . Calculate the percentage decrease in heat loss if thickness of the brick-work is increased by 100%. The ambient temperature is 30°C and assume that thermal conductivity of brick work and convective heat transfer co-efficient remain the same before and after increase in thickness. [P.U. 2009 Fall]

Soln:-



→ | M | M |
B t_1 k_1 t_2 k_2 t_3



→ | M | M |
B t_1 k_1 t_2 k_2 t_3

Case (I) when the slab thickness is x

$$\dot{Q} = \frac{t_1 - t_2}{R_1} = \frac{t_2 - t_\infty}{R_2} = \frac{t_1 - t_\infty}{R_1 + R_2}$$

$$\therefore \dot{Q} = \frac{650 - 225}{\frac{x}{kA}} = \frac{225 - 30}{\frac{1}{hA}} \quad \dots (i)$$

case (II) :- when the slab thickness is increased by 100%.

we have,

$$\dot{Q}' = \frac{t_1 - t_\infty}{\frac{2x}{kA} + \frac{1}{hA}} \quad \dots (ii)$$

From eqn (i) we have,

$$\frac{1}{hA} (650 - 225) = \frac{x}{kA} (225 - 30)$$

$$\therefore \frac{1}{hA} \times 425 = \frac{x}{kA} \times 195$$

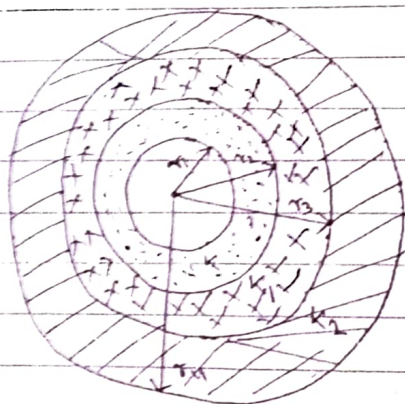
$$\frac{1}{hA} = \frac{x}{kA} \left(\frac{195}{425} \right)$$

$$= 0.459 \frac{x}{kA} \quad \dots (iii)$$

putting the value of $\frac{1}{hA}$ in eqn (2), we get,

(5) A steam pipe 13 cm inside diameter and 20 cm outside diameter is covered with two layers of insulation. The thickness of the first and the second layers is 4 cm and 8 cm respectively and their corresponding thermal conductivities are 0.15 and 0.08 W/mK respectively. The conductivity of the pipe material is 50 W/mK. The temperature of the inner surface of the steam pipe is 350°C and that of the outer surface of insulation is 30°C . Find the quantity of heat lost per meter length of the pipe. Also find out the percentage decrease in the heat loss with two layers of insulation compared to only first layer of insulation. [P.V. 2005 Fall].

Solⁿ:-



$$T_i = 350^{\circ}\text{C}$$

$$T_o = 30^{\circ}\text{C}$$

conductivity of the pipe (k) = 50 W/mK

$$k_1 = 0.15 \text{ W/mK}$$

$$k_2 = 0.08$$

$$r_1 = 0.09 \text{ m}$$

$$r_2 = 0.1 \text{ m}$$

$$r_3 = 0.1 + 0.04 = 0.14$$

$$r_4 = 0.14 + 0.08 = 0.22$$

$$R = 2\pi l (T_i - T_o)$$

$$\frac{\ln\left(\frac{r_2}{r_1}\right)}{K} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{K_1} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{K_2}$$

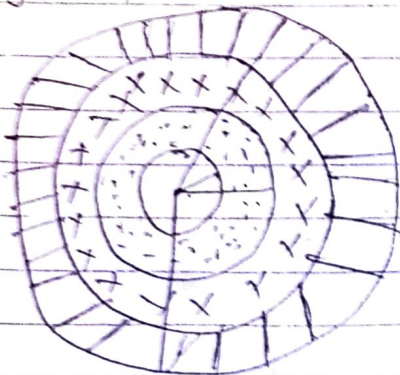
$$= 2\pi l (350 - 30)$$

$$\frac{\ln(0.1/0.09)}{50} + \frac{\ln(0.14/0.15)}{0.15} + \frac{\ln(0.22/0.14)}{0.08}$$

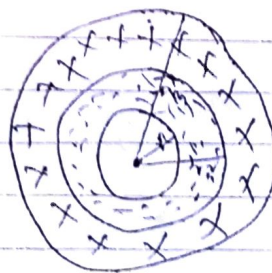
$$\begin{aligned} \therefore \frac{R}{l} &= \frac{2\pi \times 320}{0.0021 + 2.243 + 5.649} \\ &= \frac{2\pi \times 320}{7.894} \\ &= 254.7 \text{ W/m surf.} \end{aligned}$$

$$\therefore \frac{R}{l} = 254.7 \text{ W/m surf.}$$

Again,



Case - I



Case - II

Now, case-I

$$Q_1 = \frac{2\pi l (350 - 30)}{0.0021 + 2.243 + 5.649}$$
$$= \frac{2\pi l \times 320}{7.894} \quad \text{--- (i)}$$

case-II

$$Q_2 = \frac{2\pi l \times 320}{0.0021 + 2.243}$$
$$= \frac{2\pi l \times 320}{2.245} \quad \text{--- (ii)}$$

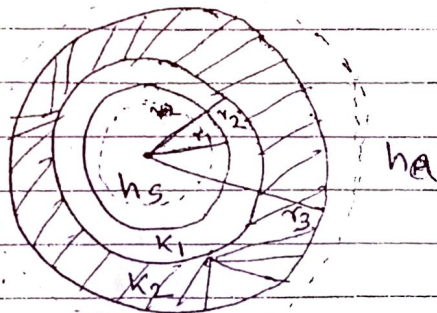
From eqn (i) & (ii).

$$\frac{Q_1}{Q_2} = 0.284$$

Percentage decrease in the heat loss with two layer of insulation compared to only first layer of insulation = $\left(1 - \frac{Q_1}{Q_2}\right) \times 100$

$$= (1 - 0.284) \times 100$$
$$= 71.6 \% \text{ Ans}$$

A 150 mm steam pipe has inside diameter of 120 mm and outside diameter of 160 mm. It is insulated at the outside with asbestos. The steam temperature is 150°C and the air temperature is 20°C , h (steam side) $= 100 \text{ W/m}^2\text{K}$, h (air side) $= 30 \text{ W/m}^2\text{K}$, k (asbestos) $= 0.5 \text{ W/mK}$, and k (steel) $= 42 \text{ W/mK}$. How thick should be asbestos be provided in order to limit the heat loss to 2.1 kW/m^2 ?



Given that,

$$r_1 = 0.06 \text{ m}$$

$$r_2 = 0.08 \text{ m}$$

$$k_1 = 42 \text{ W/mK}$$

$$k_2 = 0.5 \text{ W/mK}$$

$$h_s = 100 \text{ W/m}^2\text{K}$$

$$h_a = 30 \text{ W/m}^2\text{K}$$

$$t_s = 150^{\circ}\text{C}$$

$$t_a = 20^{\circ}\text{C}$$

$$\text{Heat loss} = 2.1 \text{ kW/m}^2$$

thickness of insulation (asbestos),
 $(r_3 - r_2) \delta$

Area for heat transfer = $2\pi r L$ (where L = length of the pipe)

$$\begin{aligned}\therefore \text{Heat loss} &= 2.1 \times 2\pi r L \\ &= 2.1 \times 2\pi \times 0.075 \times L \\ \therefore Q &= 0.989 L \text{ kW}\end{aligned}$$

where, r , mean radius = $\frac{150}{2} = 75 \text{ mm}$
 $\text{or } 0.075 \text{ m. inner}$

Heat transfer rate in such a case is given by,

$$Q = \frac{2\pi L (t_s - t_a)}{\left[\frac{1}{h_s \cdot r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_a \cdot r_3} \right]}$$

$$\begin{aligned}\S \quad 0.989 L \times 10^3 &= \frac{2\pi L (150 - 20)}{\left[\frac{1}{100 \times 0.06} + \frac{\ln(0.08/0.06)}{42} + \frac{\ln(r_3/0.08)}{0.8} + \frac{1}{30 \times r_3} \right]}\end{aligned}$$

$$\begin{aligned}\S \quad 0.989 \times 10^3 &= \frac{816.81}{\left[0.1666 + 0.0068 + \frac{\ln(r_3/0.08)}{0.8} + \frac{1}{30 \times r_3} \right]}\end{aligned}$$

$$\frac{\ln(r_3/0.08)}{0.8} + \frac{1}{30r_3} = \frac{816.81 - (10.66 + 0.125)}{0.089}$$

$$\S \frac{1.25 \ln(r_3/0.08)}{30r_3} - 0.652 = 0$$

$$\S 1.25 \ln(r_3/0.08) = 0.652 \times 30r_3$$

$$\S 1.25 (\ln r_3 - \ln 0.08) = 19.56 r_3$$

$$\S \ln r_3 - \ln 0.08 = \frac{19.56 r_3}{1.25}$$

$$\S 1 + 2.525 = 15.648 r_3$$

solving by hit and trial method, we get,

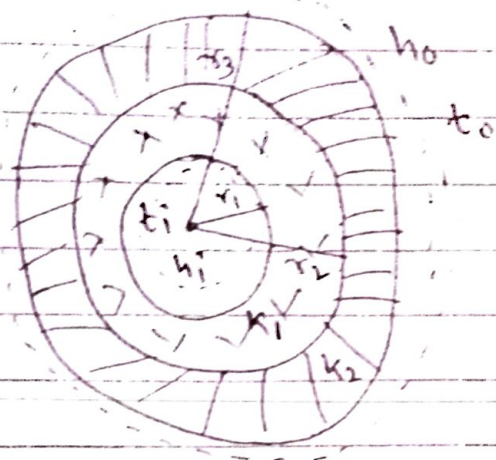
$$r_3 = 0.105 \text{ m} \\ = 105 \text{ mm}$$

$$\therefore \text{Thickness of insulation} = r_3 - r_2 \\ = 105 - 80 \\ = 25 \text{ mm Ans}$$

- (7) Hot air at a temperature of 60°C is blowing through a steel pipe of 10 cm diameter. The pipe is covered with two layers of different insulating materials of thickness 5 cm and 3 cm, and their corresponding thermal conductivities are 0.23 and 0.37 W/mK. The inside and outside heat transfer coefficients are 58 and 12 W/m²K. The atmosphere is at

25°C Find the rate of heat loss from a 50m long pipe. Neglect the resistance of the steel pipe.

Soln:-



$$r_1 = 0.05$$

$$r_2 = 0.05 + 0.05 = \cancel{0.08} 0.1$$

$$r_3 = \cancel{0.08} + 0.1 + 0.03 = 0.13$$

$$l = 50\text{m}$$

$$k_1 = 0.23 \text{ W/mK}$$

$$k_2 = 0.37 \text{ W/mK}$$

$$h_i = 58 \text{ W/m}^2\text{K}$$

$$h_o = 12 \text{ W/m}^2\text{K}$$

$$t_i = 60^\circ\text{C}$$

$$t_o = 25^\circ\text{C}$$

$\therefore$ Rate of heat loss,

$$Q = \frac{2\pi l (t_i - t_o)}{\left[\frac{1}{h_i r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_o r_3} \right]}$$

$$= 2\pi \times 50 (60 - 25)$$

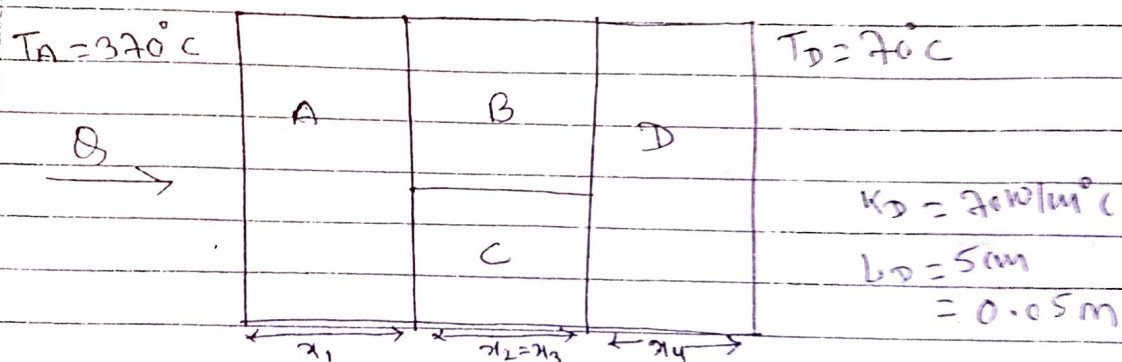
$$\left[\frac{1}{58 \times 0.05} + \frac{\ln(0.1/0.05)}{0.23} + \frac{\ln(0.2/0.1)}{0.37} + \frac{1}{1210 \times 3} \right]$$

$$= 2\pi \times 50 \times 35$$

$$0.344 + 3.01 + 0.709 + 0.277$$

$$= 2.533 \text{ kW } \underline{\text{Ans}}$$

(8) Find the heat transfer through the composite wall shown in figure [DU. 2009 spring]



$$k_A = 150 \text{ W/m}^\circ\text{C}; \quad l_A = 2.5 \text{ cm} = 0.025 \text{ m}$$

$$k_B = 30 \text{ W/m}^\circ\text{C}; \quad l_B = 7.5 \text{ cm} = 0.075 \text{ m}$$

$$k_C = 50 \text{ W/m}^\circ\text{C}; \quad l_C = 7.5 \text{ cm} = 0.075 \text{ m}$$

$$A_A = A_D = 2A_B = 2A_C = 0.2 \text{ m}^2$$

$$\underline{\text{Sol}^n} \quad x_A = \frac{0.2}{0.025} = 8$$

$$x_B = x_C = \frac{0.1}{0.075} = 1.33$$

$$\gamma_D = \frac{0.2}{0.05} = 4$$

$$B = \frac{I_A}{K_A A}$$

$$R_A = \frac{\gamma_A}{K_A A_A} = \frac{0.025}{150 \times 0.2} = 0.267 \times 10^{-3}$$

$$R_B = \frac{\gamma_B}{K_B A_B} = \frac{0.075}{30 \times 0.1} = 0.443 \times 10^{-3}$$

$$R_C = \frac{\gamma_C}{K_C A_C} = \frac{0.075}{50 \times 0.1} = 0.266 \times 10^{-3}$$

$$R_D = \frac{\gamma_D}{K_D A_D} = \frac{0.05}{70 \times 0.2} = 0.285 \times 10^{-3}$$

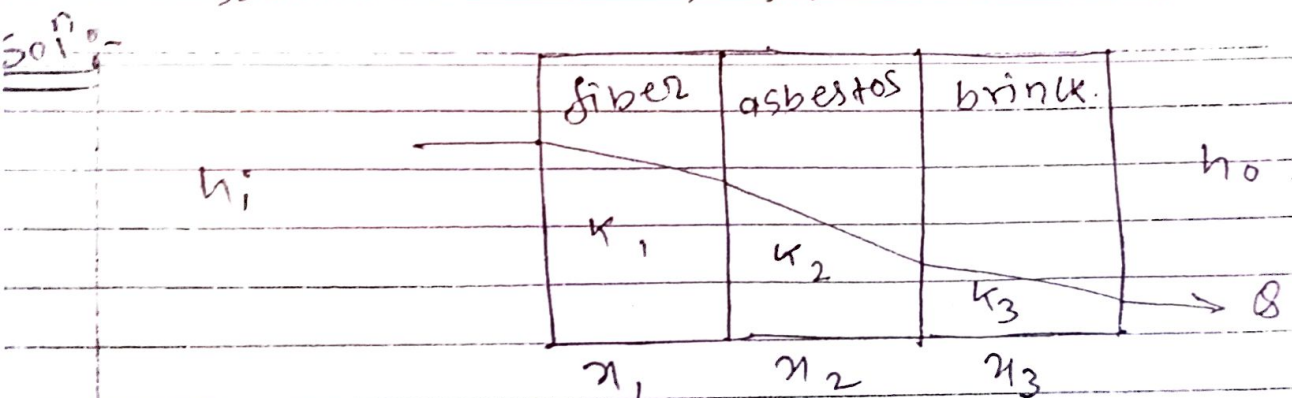
$$R_{eq} = \frac{R_B \times R_C}{R_B + R_C}$$

$$= \frac{0.443 \times 0.266}{0.443 + 0.266} \times 10^{-3} = 0.37 \times 10^{-3}$$

$$\therefore \theta = \frac{\tau_A - \tau_D}{R_A + R_{eq} + R_D}$$

$$= \frac{0.267 + 0.166 + 0.287}{8.33 \times 10^{-4} + 0.37 \times 10^{-3} + 3.5 \times 10^{-4}} = 21.7 \text{ kN/mm}^2$$

(9) A house wall may be approximated as two layers of fiber insulating board, a 3cm layer of loosely packed asbestos and a 10cm layer of common brick assuming convection heat transfer co-efficient of $15 \text{ W/m}^2\text{K}$ on both sides of the wall, calculate the overall heat transfer coefficient for this arrangement $[k_{\text{fib}} = 0.7 \text{ W/mK}; k_{\text{asbestos}} = 0.18 \text{ W/mK}; k_{\text{brick}} = 0.7 \text{ W/mK}]$



$$x_1 = 0.24 \text{ m}$$

$$x_2 = 0.08 \text{ m}$$

$$x_3 = 0.1 \text{ m}$$

$$k_1 = 0.7$$

$$k_2 = 0.18$$

$$k_3 = 0.7$$

$$h_i = h_o = 15$$

overall heat transfer co-efficient $(U) = ?$

$$\frac{1}{U} = \frac{1}{15} + \frac{0.24}{0.7} + \frac{0.08}{0.18} + \frac{0.1}{0.7} + \frac{1}{15}$$

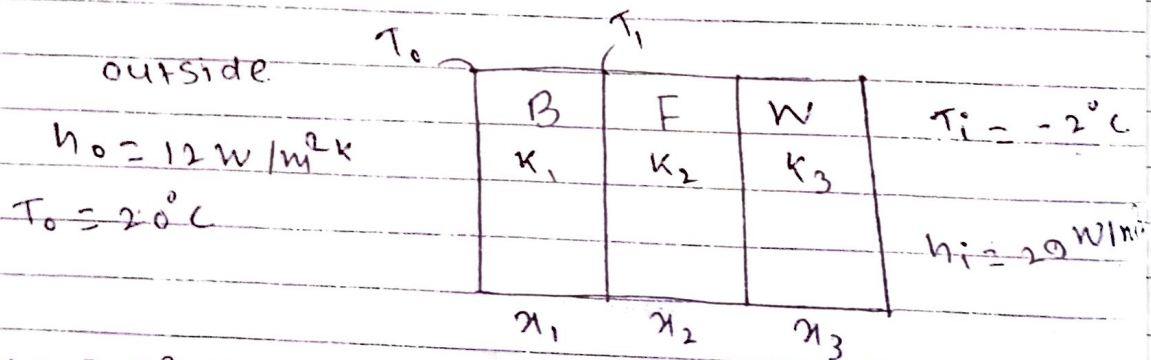
$$\frac{1}{U} = 1.063$$

$$\therefore U = 0.94 \text{ W/m}^2\text{K}$$

inside surface की अपेक्षा कम है।

Q.1) A cold storage room has walls made of 0.23 m of brick on the outside, 0.08 m of plastic foam and finally 1.5 cm of wood on the inside. The outside and inside air temperatures are 20°C and -2°C respectively. If the inside and outside heat transfer coefficients are respectively $20, 12 \text{ W/m}^2\text{K}$ and the thermal conductivities of brick, foam and wood are $0.98, 0.02$ and 0.17 W/mK respectively, determine (i) the rate of heat removed by refrigeration if the total wall area is 90 m^2 , and (ii) temperature of the inside surface of the brick.

Solⁿ:-



$$\begin{aligned} x_1 &= 0.23 \text{ m} & k_1 &= 0.98 \\ x_2 &= 0.08 \text{ m} & k_2 &= 0.02 \\ x_3 &= 0.015 \text{ m} & k_3 &= 0.17 \end{aligned}$$

$$\theta = T_0 - T_i$$

$$\frac{1}{h_0 A} + \frac{x_1}{k_1 A} + \frac{x_2}{k_2 A} + \frac{x_3}{k_3 A} + \frac{1}{h_i A}$$

$$= \frac{A(T_0 - T_i)}{A \left(\frac{1}{h_0} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{1}{h_i} \right)}$$

$$\frac{1}{h_0} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3} + \frac{1}{h_i}$$

$$= 90 \times 22$$

$$= 8.33 \times 10^{-2} + 2.35 \times 10^{-1} + 4 + 8.33 \times 10^{-2} + 3.45 \times 10^{-2}$$

$$\therefore \dot{Q} = 446.33 \text{ W} \cdot \text{m}^2$$

Again,

$$\dot{Q} = \frac{T_0 - T_1}{\frac{1}{hA} + \frac{x_1}{k_1 A}}$$

$$A = 90$$

$$\text{total breadth} = 0.325$$

$$\text{length} = 276.92$$

$$\text{Area of B} = 276.92 \times 0.23 = 63.69$$

$$446.33 = \frac{A(20 - T_1)}{(8.33 \times 10^{-2} + 2.35 \times 10^{-1})}$$

$$\S 446.33 \times 0.318 = A(20 - T_1)$$

$$\S 141.93 = 63.69(20 - T_1)$$

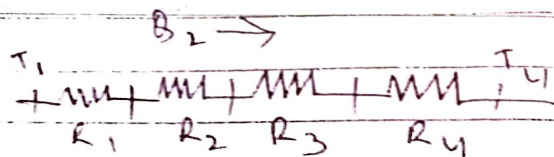
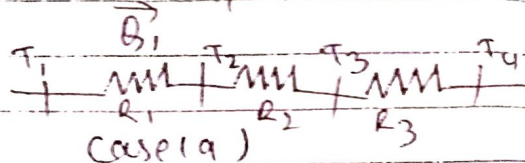
$$\S 141.93 - 1273.8 = -63.69 T_1$$

$$\therefore T_1 = 17.77^\circ \text{C} \quad \text{Ans}$$

(II) A furnace wall is made of steel plate 10mm thick ($k = 62.8 \text{ kJ/m-hr-}^\circ\text{C}$) lined on inside with silica bricks 150mm thick ($k = 7.32 \text{ kJ/m-hr-}^\circ\text{C}$) and on the outside with magnesia bricks 200mm thick ($k = 18.84 \text{ kJ/m-hr-}^\circ\text{C}$). The inside and outside surfaces of the wall are at temperature 650°C and 125°C respectively. Make calculations for the heat loss from unit area of the wall.

It is required that the heat loss be reduced to 10 MJ/hr by means of air gap between steel and magnesia bricks. Estimate the necessary width of air gap if the thermal conductivity for air is $0.126 \text{ kJ/m-hr-}^\circ\text{C}$.

Sol:



Thermal resistance for a plane wall of thickness δ , area A and thermal conductivity k is prescribed by the relation $R = \delta / kA$.

$\therefore$ Resistance of steel plate,

$$R_1 = \frac{0.01}{62.8 \times 1} = 0.000159 \frac{\text{deg hr}}{\text{kJ}}$$

Resistance of bricks,

$$R_2 = \frac{0.15}{7.32 \times 1} = 0.02049 \frac{\text{deg hr}}{\text{kJ}}$$

Resistance of magnesia bricks,

$$R_3 = \frac{0.2}{18.84 \times 1} = 0.01061 \frac{\text{deg hr}}{\text{kJ}}$$

Total resistance of the composite wall,

$$R_t = R_1 + R_2 + R_3 = 0.03126 \frac{\text{deg hr}}{\text{kJ}}$$

Heat loss from the wall,

$$= \frac{\Delta t}{R_t} = \frac{650 - 125}{0.03126}$$

$$= 16795 \text{ kJ/hr}$$

Again

To reduce the heat loss to 10 MJ/hr , the thermal resistance should be increased to:

$$\frac{650 - 125}{10 \times 10^3} = 0.0525 \frac{\text{deg hr}}{\text{kJ}}$$

∴ Thermal resistance for the air gap,

$$\begin{aligned} \frac{\delta}{kA} &= 0.0525 - 0.03126 \\ &= 0.02124 \text{ deg hr/kJ} \end{aligned}$$

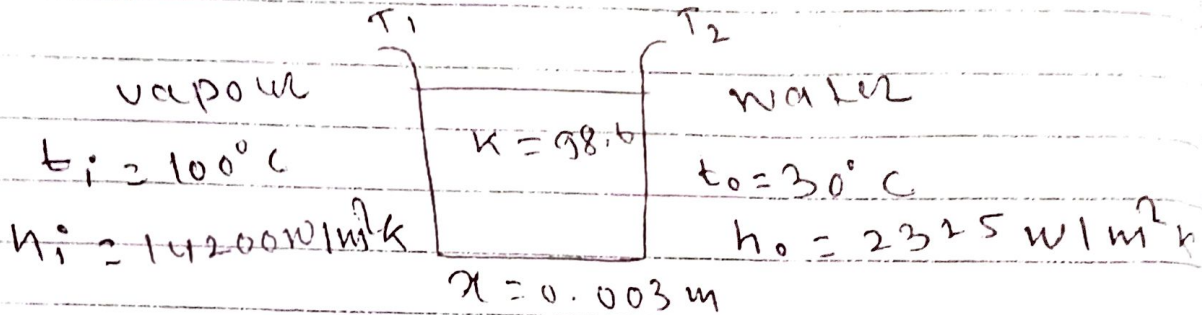
∴ Thickness of air gap,

$$\begin{aligned} \delta &= 0.02124 \times 0.126 \times 1 \\ &= 2.676 \times 10^{-3} \text{ m} \\ &= 2.676 \text{ mm} \end{aligned}$$

- 13) A 3mm thick metal plate having thermal conductivity $k = 98.6 \text{ W/mK}$, is exposed to vapour at 100°C on one side and cooling water at 30°C on the opposite side. The heat transfer co-efficient are: $h_i = 14200 \text{ W/m}^2\text{K}$ on the vapour side; $h_o = 2325 \text{ W/m}^2\text{K}$ on the water side. Determine the rate of heat

transfer, the overall heat transfer coefficient and the drop in temperature at each side of heat transfer.

Soln:-



$$\dot{Q} = \frac{A (t_i - t_o)}{\frac{1}{h_i} + \frac{x}{k} + \frac{1}{h_o}}$$

$$\frac{\dot{Q}}{A} = \frac{100 - 30}{\frac{1}{14200} + \frac{0.003}{98.6} + \frac{1}{2325}}$$

$$= 70$$

$$= \frac{7.04 \times 10^{-5} + 3.042 \times 10^{-5} + 4.3 \times 10^{-4}}{}$$

$$= 131871.44$$

$$= 1.31 \times 10^5 \text{ W/m. Ans}$$

Again,

$$U = \frac{1}{\frac{1}{h_o} + \frac{x}{k} + \frac{1}{h_i}}$$

$$= \frac{1}{}$$

$$\frac{1}{\frac{1}{14200} + \frac{0.003}{98.6} + \frac{1}{2325}}$$

$$= 1883.87$$

Ans

(11) Saturated steam at 110°C flows inside a copper pipe (thermal conductivity 450 W/mK) having an internal diameter of 10 cm and an external diameter of 12 cm . The surface resistance on the steam side is $12000 \text{ W/m}^2\text{K}$ and that on the outside surface of pipe is $18 \text{ W/m}^2\text{K}$. Determine the heat loss from the pipe if it is located in space at 25°C . How this heat loss would be affected if the pipe is lagged with 5 cm thick insulation of thermal conductivity 0.22 W/mK .

Solⁿ:-

$$T_1 = 110^\circ\text{C}$$

$$k_1 = 450 \text{ W/mK}$$

$$r_1 = 0.05 \text{ m}$$

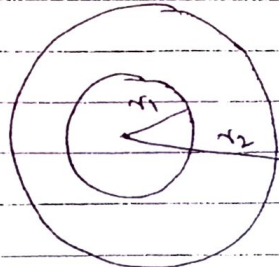
$$r_2 = 0.06 \text{ m}$$

$$r_3 = 0.11$$

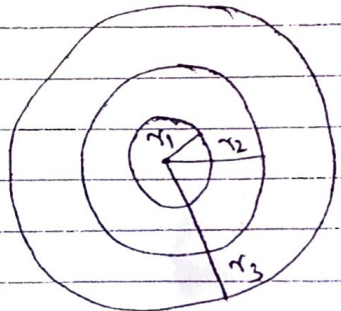
$$h_1 = 12000 \text{ W/m}^2\text{K}$$

$$T_2 = 25^\circ\text{C}$$

$$k_2 = 0.22 \text{ W/mK}$$



Case - I



Case - II

$$\dot{Q}_1 = \frac{2\pi (T_1 - T_2)}{\frac{1}{h_1 r_1} + \frac{\ln(r_2/r_1)}{k} + \frac{1}{h_2 r_2}}$$

$$= 2\pi \times 85$$

$$\frac{1}{0.05 \times 12000} + \frac{\ln(0.06/0.05)}{450} + \frac{1}{0.06 \times 18}$$

$$= 2\pi \times 85$$

$$1.67 \times 10^{-3} + 4.05 \times 10^{-4} + 0.26 \times 10^{-1}$$

$$= 575.46 \text{ W/m}$$

Again,

$$\frac{Q_2}{L} = \frac{2\pi(T_1 - T_2)}{\frac{1}{h_1 r_1} + \frac{\ln(r_2/r_1)}{k} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_2 r_3}}$$

$$= \frac{2\pi \times 85}{\frac{1}{0.05 \times 12000} + \frac{\ln(0.06/0.05)}{450} + \frac{\ln(0.11/0.06)}{450} + \frac{1}{0.11 \times 18}}$$

$$= \frac{2\pi \times 85}{1.67 \times 10^{-3} + 4.05 \times 10^{-4} + 2.75 + 0.5}$$

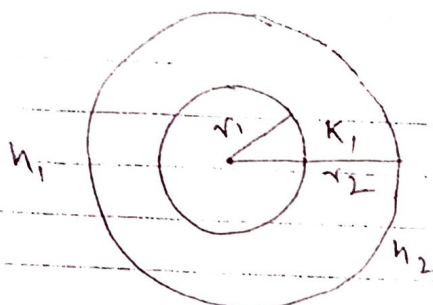
$$= 164.22$$

$$\% \text{ heat loss} = \frac{(575.46 - 164.22)}{575.46} \times 100\%$$

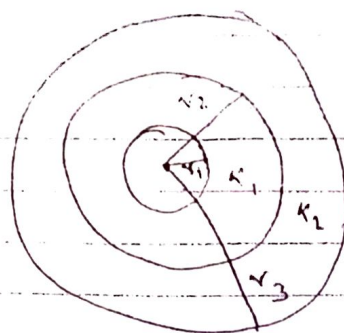
$$= 71.46\% \text{ Ans}$$

- (15) An insulated steam pipe of 16 cm diameter is covered with a cm layer of insulation ($k = 0.9 \text{ W/mK}$) and carries process steam. Determine the percentage change in the rate of heat loss if extra 2 cm thick layer of lagging ($k = 1.25 \text{ W/mK}$) is provided. Given that surrounding temperature remains constant and the heat transfer coefficient for both configurations is $12 \text{ W/m}^2\text{K}$.

Solⁿ



Case - I



Case - II

$$r_1 = 0.08 \text{ m}$$

$$r_2 = 0.12 \text{ m}$$

$$r_3 = 0.14 \text{ m}$$

$$k_1 = 0.9 \text{ W/mK}$$

$$k_2 = 1.25 \text{ W/mK}$$

$$h_1 = h_2 = 12 \text{ W/m}^2\text{K}$$

$$\begin{aligned} \frac{Q}{l} &= \frac{2\pi T}{\frac{1}{h_1 r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{1}{h_1 r_2}} \\ &= \frac{2\pi T}{\frac{1}{12 \times 0.08} + \frac{\ln(0.12/0.08)}{0.9} + \frac{1}{0.12 \times 12}} \\ &= 0.457 T \times 2\pi \end{aligned}$$

And,

$$\begin{aligned} \frac{Q}{l} &= \frac{2\pi T}{\frac{1}{h_1 r_1} + \frac{\ln(r_2/r_1)}{k_1} + \frac{\ln(r_3/r_2)}{k_2} + \frac{1}{h_1 r_3}} \\ &= \frac{2\pi T}{\frac{1}{12 \times 0.08} + \frac{\ln(0.12/0.08)}{0.9} + \frac{\ln(0.14/0.12)}{1.25} + \frac{1}{12 \times 0.14}} \end{aligned}$$

$$= 0.452 T$$

$$\begin{aligned} \% \text{ change in heat loss} &= \frac{2\pi \times 4.57 T - 0.452 T \times 2\pi}{2\pi (0.457 T)} \times 100 \% \\ &= 1.09 \% \end{aligned}$$

(16) A steam main 75 mm inside diameter and given outside diameter is lagged with two successive layers of insulations. The layer in contact with the pipe is 38 mm asbestos and the asbestos layer is covered with 25 mm thick magnesia insulation. The surface co-efficient for inside and outside surfaces are $22.7 \text{ W/m}^2\text{K}$ and $6.8 \text{ W/m}^2\text{K}$ respectively. If the steam temperature is 375°C and the ambient temperature is 35°C , calculate the steady state loss of heat from steam per 60 m length of pipe. Also work out the overall heat transfer co-efficient based on the inside and outside surfaces of the lagged steam main. [k (pipe material) = 45 W/mK ; k (asbestos) = 0.14 W/mK ; k (magnesia insulation) = 0.07 W/mK]

Sol:

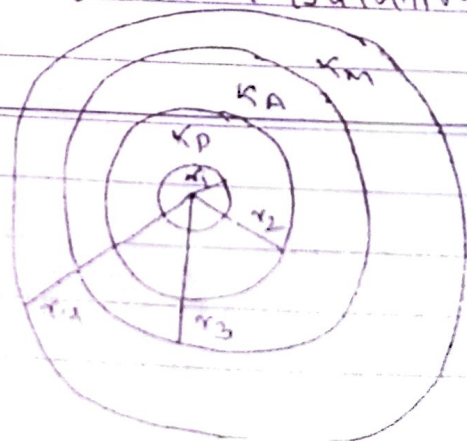
$$t_s = 375^\circ\text{C}$$

$$t_a = 35^\circ\text{C}$$

$$l = 60 \text{ m}$$

$$r_1 = 0.0375$$

$$r_2 = 0.045$$



$$r_3 = 0.083$$

$$r_4 = 0.108$$

$$h_i = 227 \text{ W/m}^2\text{K}$$

$$h_o = 6.8 \text{ W/m}^2\text{K}$$

$$k_{\text{pipe}} = 45 \text{ W/mK}$$

$$k_A = 0.14 \text{ W/mK}$$

$$k_m = 0.07 \text{ W/mK}$$

$$Q = \frac{2\pi L (T_g - T_\infty)}{\frac{1}{h r_1} + \frac{\ln(r_2/r_1)}{k_p} + \frac{\ln(r_3/r_2)}{k_A} + \frac{\ln(r_4/r_3)}{k_m} + \frac{1}{h r_4}}$$

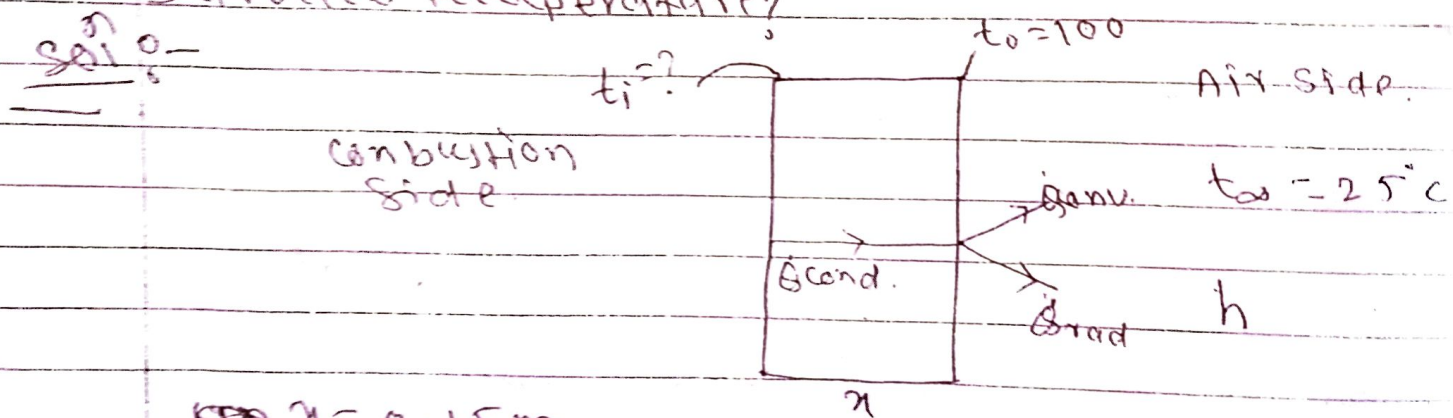
$$= 2\pi \times 60 (375 - 35)$$

$$\frac{1}{227 \times 0.0375} + \frac{\ln(0.045/0.0375)}{45} + \frac{\ln(0.083/0.04)}{0.14} + \frac{\ln(0.108/0.083)}{0.07} + \frac{1}{6.8 \times 0.108}$$

$$= \frac{340 \times 2\pi \times 60}{1.17 \times 10^{-1} + 4.05 \times 10^{-3} + 4.37 + 3.76 + 1.31}$$

$$= 13331.4 \text{ W} \quad \underline{\underline{\text{Ans}}}$$

Q7) The hot combustion gases of a furnace are separated from the ambient air and its surroundings, which are at 25°C , by a brick wall 0.15 m thick. The brick has a thermal conductivity of 1.5 W/mK and a surface emissivity of 0.8 . Under steady state conditions, an outer surface temperature of 100°C is measured. Free convection heat transfer to the air adjoining the surface is characterized by a convection coefficient of $h = 20\text{ W/m}^2\text{K}$. What is the brick inner surface temperature?



$x = 0.15\text{ m}$
 $k = 1.5\text{ W/mK}$
 $\epsilon = 0.8$
 $t_o = 100^\circ\text{C}$
 $t_i = ?$

Under steady state,
 $Q_{cond} = Q_{conv} + Q_{rad}$

$$\frac{kA(t_i - t_o)}{x} = hA(t_o - T_{\infty}) + \epsilon \sigma A(T_o^4 - T_{\infty}^4)$$

$$Q = \frac{2\pi r L (T_1 - T_2)}{\ln(r_2/r_1)}$$

$$T_1 - T_2 = \frac{2L}{k} [h(T_2 - T_\infty) + \epsilon \sigma (T_2^4 - T_\infty^4)]$$

$$\sum T_1 = \frac{2L}{k} [h(T_2 - T_\infty) + \epsilon \sigma (T_2^4 - T_\infty^4)] + T_2$$

$$= \frac{0.15}{1.05} [20(100 - 25) + 0.8 \times 5.67 \times 10^{-8} (100^4 - 125^4)]$$

$$+ T_2$$

$$= 150.45 + 100$$

$$= 250.45^\circ \text{C} \text{ Ans}$$

(13) A lake surface is covered by a 8 cm thick layer of ice [$k = 2.23 \text{ W/mK}$] when the ambient air temperature is -12.5°C . A thermocouple embedded on the upper surface of the ice layer indicates a temperature of -5°C . Assuming steady state condition in ice and no liquid subcooling at the bottom surface of the ice layer, Find the heat transfer coefficient at the upper surface. Also find out the heat loss per unit area.

Solⁿ:-

$$k_i = 2.23 \text{ W/mK}$$

$$T_\infty = -12.5^\circ \text{C}$$

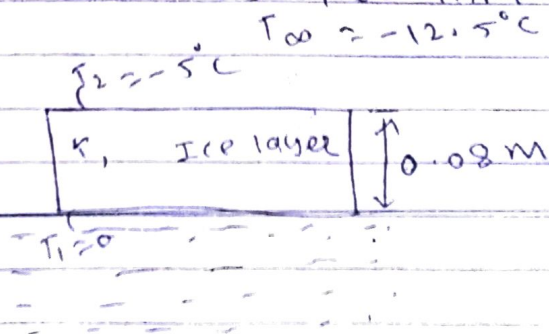
$$T_2 = -5^\circ \text{C}$$

$$T_1 = 0$$

$$x = 0.08$$

$$h = ?$$

$$\dot{Q} = ?$$



$$Q = \frac{kA(T_1 - T_2)}{x}$$

$$\frac{Q}{A} = \frac{2.23(0+5)}{0.08}$$

$$= 139.37 \text{ W/m}^2 \text{ Ans.}$$

For steady condⁿ.

$$Q = hA(T_2 - T_1)$$

$$\{ 139.37 = h(-5+12.5)$$

$$\{ h = \frac{139.37}{7.5}$$

$$= 18.58 \text{ W/m}^2\text{K Ans.}$$

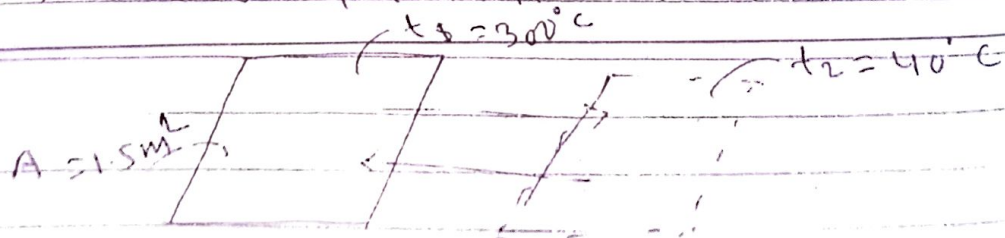
(19) A surface having an area of 1.5 m^2 and maintained at 300°C exchanges heat by radiation with another surface at 40°C . The value of factor due to the geometric location and emissivity is 0.52. determine:

(a) Heat lost by radiation.

(b) The value of thermal resistance.

(c) The value of equivalent convection coefficient.

Sol:-



Given that

$$f_{12} = 0.52$$

$$T_1 = 300 + 273 = 573 \text{ K}$$

$$T_2 = 40 + 273 = 313 \text{ K}$$

$$A = 1.5 \text{ m}^2$$

(a) Net radiation

$$\dot{Q} = \sigma A f_{12} (T_1^4 - T_2^4)$$

$$= 5.67 \times 10^{-8} \times 1.5 \times 0.52 (573^4 - 313^4)$$

$$= 4343.08 \text{ W} \quad \text{Ans}$$

(b)

$$\dot{Q} = \frac{(T_1 - T_2)}{R}$$

$$\therefore R = \frac{(T_1 - T_2)}{\dot{Q}}$$

$$= \frac{573 - 313}{4343.08}$$

$$= 0.0598 \text{ } ^\circ\text{K/W} \quad \text{Ans}$$

(c)

$$\dot{Q} = hA(T_1 - T_2)$$

$$h = \frac{\dot{Q}}{A(T_1 - T_2)} = \frac{4343.08}{1.5 \times (573 - 313)} = 11.136 \text{ W/m}^2\text{ } ^\circ\text{K} \quad \text{Ans}$$

- (22) A cylindrical rod, 1.5 m long and 2 cm in diameter, is heated electrically and positioned in a vacuum furnace which has interior walls at 800 K temperature. A controlled amount of current is passed through the rod and its surface is maintained at 1000 K. Calculate the power supplied to the heating rod if its surface has ~~emissivity~~ $\epsilon = 0.9$.

Soln:

$$l = 1.5 \text{ m}$$

$$d = 0.02 \text{ m}$$

$$T_1 = 1000 \text{ K}$$

$$T_2 = 800 \text{ K}$$

$$\epsilon = 0.9$$

$$\sigma = 5.67 \times 10^{-8}$$

$$P = \sigma A \epsilon (T_1^4 - T_2^4)$$

$$= \sigma (\pi d l) \epsilon (T_1^4 - T_2^4)$$

$$= 5.67 \times 10^{-8} \times (\pi \times 0.02 \times 1.5) \times 0.9 (1000^4 - 800^4)$$

$$= 2839.5 \text{ W} \text{ Ans .}$$

- (23) Two opposed, parallel infinite planes are maintained at 420 K and 480 K respectively. Calculate the net heat flux between these planes if one has an emissivity of 0.8 and other has 0.7. Does it matter

which plate has which emissivity. How the heat flux will be affected i.e.

(a) The temperature difference is doubled by raising 480K to 540K.

(b) The plates are assumed to be black.

Solⁿ:-

Incident 11-plate

$$f_{12} = \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$= \frac{1}{\frac{1}{0.8} + \frac{1}{0.7} - 1}$$

$$f_{12} = 0.59$$

$$T_1 = 420K$$

$$T_2 = 480K$$

$$\epsilon_1 = 0.8$$

$$\epsilon_2 = 0.7$$

heat flux,

$$\left(\frac{Q_{21}}{A} \right) = \sigma f_{12} (T_2^4 - T_1^4)$$

$$= 5.67 \times 10^{-8} \times 0.59 ((480)^4 - (420)^4)$$

$$= 734.86 \text{ W/m}^2$$

(a) When the temperature difference is double,

heat flux.

$$T_2 = 540, T_1 = 420$$

$$\left(\frac{Q_{21}}{A} \right) = \sigma f_{12} (T_2^4 - T_1^4)$$

$$= 5.67 \times 10^{-8} \times 0.59 ((540)^4 - (420)^4)$$

$$= 1803.55 \text{ W/m}^2$$

$$\text{Heat flux increase} = \frac{1803.55}{734.86}$$

$\therefore 2.45 \text{ times}$

(b) If the plates are black,

$$\epsilon_1 = \epsilon_2$$

$$f = 1$$

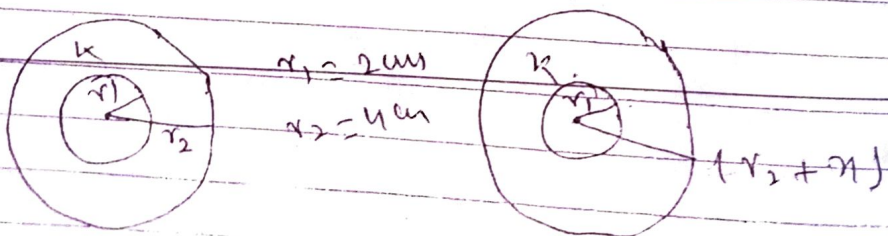
$$\left[\frac{Q_{21}}{A} \right] = \sigma (T_2^4 - T_1^4)$$

$$= 5.67 \times 10^{-8} ((480)^4 - (420)^4)$$

$$= 1245.54 \text{ W/m}^2 \text{ Ans.}$$

(24) A hot fluid is being conveyed through a long pipe of 4 cm outer diameter and covered with 2 cm thick insulation. It is proposed to reduce the conduction heat loss to the surroundings to one third of the present rate by further covering with same insulation. Calculate the additional thickness of the insulation.

Soln.



Case I -

Case I - I

$$Q_1 = \frac{2\pi k l (T_1 - T_2)}{\ln(r_2/r_1)}$$

Case II - II

$$Q_2 = \frac{2\pi k l (T_1 - T_2)}{\ln(r_2 + x/r_1)}$$

According to question,

$$Q_2 = Q_1/3$$

$$\Rightarrow \frac{2\pi k l (T_1 - T_2)}{\ln(r_2 + x/r_1)} = \frac{1}{3} \frac{2\pi k l (T_1 - T_2)}{\ln(r_2/r_1)}$$

$$\Rightarrow \ln\left(\frac{r_2 + x}{r_1}\right) = 3 \ln\left(\frac{r_2}{r_1}\right)$$

$$\Rightarrow \frac{r_2 + x}{r_1} = \left(\frac{r_2}{r_1}\right)^3$$

$$\Rightarrow \frac{4 + x}{2} = \left(\frac{4}{2}\right)^3$$

$$\Rightarrow \frac{4 + x}{2} = 8$$

$$\Rightarrow x = 16 - 4$$

$$\therefore x = 12 \text{ cm}$$

Ans

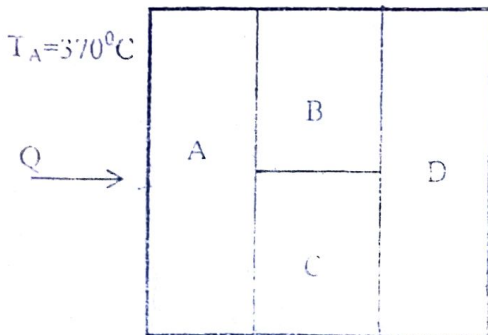
Prabin chaudhary

NAST

BE Computer II
(2022 spring)

Heat Transfer

1. A brick wall 450 mm thick is plastered with concrete 30 mm thick. The thermal conductivity of the brick and the concrete are 0.7 W/mK and 0.92 W/mK , respectively. If the temperature of the exposed brick face is 35°C and that of concrete is 10°C , determine the heat loss per hour through the wall which is 5 m long and 3.0 m high. Also determine the interface temperature. [P.U. 2006 Spring]
2. An exterior wall of a house may be approximated by a 100 mm layer of common brick ($k=0.7 \text{ W/mK}$) followed by a 40 mm layer of gypsum plaster ($k=0.48 \text{ W/mK}$). What thickness of loosely packed rock-wool insulation ($k=0.065 \text{ W/mK}$) should be added to reduce the heat loss (or gain) through the wall by 80%? [P.U. 2007 Fall]
3. The temperatures at the inside and outside surfaces of the brick-work of a furnace have been noted to be 650°C and 225°C . Calculate the percentage decrease in heat loss if thickness of the brick-work is increased by 100%. The ambient temperature is 30°C and assume that thermal conductivity of brick work and convective heat transfer co-efficient remain the same before and after increase in thickness. [P.U. 2009 Fall]
4. A 16 cm diameter pipe carrying saturated steam is covered by a layer of insulation of thickness 40 mm ($k=0.8 \text{ W/mK}$). Later on, an extra layer of insulation of 10 mm thick ($k=0.12 \text{ W/mK}$) is added. If the surrounding temperature remains constant and heat transfer coefficient (outside) for both insulation materials is $10 \text{ W/m}^2\text{K}$, determine the percentage change in the rate of heat loss due to extra insulation layer. [P.U. 2004/2008 Spring]
5. A steam pipe 18 cm inside diameter and 20 cm outside diameter is covered with two layers of insulation. The thickness of the first and the second layers is 4 cm and 8 cm respectively and their corresponding thermal conductivities are 0.15 and 0.08 W/mK respectively. The conductivity of the pipe material is 50 W/mK . The temperature of the inner surface of the steam pipe is 350°C and that of the outer surface of insulation is 30°C . Find the quantity of heat lost per metre length of the pipe. Also find out the percentage decrease in the heat loss with two layers of insulation compared to only first layer of insulation. [P.U. 2005 Fall]
6. A 150 mm steam pipe has inside diameter of 120 mm and outside diameter of 160 mm. It is insulated at the outside with asbestos. The steam temperature is 150°C and the air temperature is 20°C , $h(\text{steam side})=100 \text{ W/m}^2\text{K}$, $h(\text{air side})=30 \text{ W/m}^2\text{K}$, $k(\text{asbestos})=0.5 \text{ W/mK}$, and $k(\text{steel})=42 \text{ W/mK}$. How thick should the asbestos be provided in order to limit the heat loss to 2.1 kW/m^2 ? [P.U. 2006 Fall]
7. Hot air at a temperature of 60°C is flowing through a steel pipe of 10 cm diameter. The pipe is covered with two layers of different insulating materials of thickness 5 cm and 3 cm, and their corresponding thermal conductivities are 0.23 and 0.37 W/mK . The inside and outside heat transfer co-efficients are 58 and $12 \text{ W/m}^2\text{K}$. The atmosphere is at 25°C . Find the rate of heat loss from a 50 m long pipe. Neglect the resistance of the steel pipe. [P.U. 2007 Spring]
8. Find the heat transfer through the composite wall shown in figure. [P.U. 2009 Spring]



$$T_D = 70^\circ\text{C}$$

$$k_A = 150 \text{ W/m}^\circ\text{C};$$

$$L_A = 2.5 \text{ cm}$$

$$k_B = 30 \text{ W/m}^\circ\text{C};$$

$$L_B = 7.5 \text{ cm}$$

$$k_C = 50 \text{ W/m}^\circ\text{C};$$

$$L_C = 5 \text{ cm}$$

$$k_D = 70 \text{ W/m}^\circ\text{C};$$

$$L_D = 7.5 \text{ cm}$$

$$A_A = A_B = A_C = A_D = 1 \text{ m}^2$$

9. A house wall may be approximated as two 12 cm layers of fiber insulating board, a 8 cm layer of loosely packed asbestos, and a 10 cm layer of common brick. Assuming convection heat transfer co-efficient of $15 \text{ W/m}^2\text{K}$ on both sides of the wall, calculate the overall heat transfer co-efficient for this arrangement [$k_{\text{fib}}=0.07 \text{ W/mK}$, $k_{\text{asb}}=0.18 \text{ W/mK}$; $k_{\text{brk}}=0.7 \text{ W/mK}$].
10. A cold storage room has walls made of 0.23 m of brick on the outside, 0.08 m of plastic foam, and finally 1.5 cm of wood on the inside. The outside and inside air temperatures are 20°C and -2°C respectively. If the inside and outside heat transfer co-efficients are respectively $29, 12 \text{ W/m}^2\text{K}$ and the thermal conductivities of brick, foam and wood are $0.98, 0.02$ and 0.17 W/mK respectively, determine (i) the rate of heat removed by refrigeration if the total wall area is 90 m^2 , and (ii) the temperature of the inside surface of the brick.
11. A furnace wall is made of steel plate 10 mm thick ($k=62.8 \text{ kJ/m-hr}^\circ\text{C}$) lined on inside with silica bricks 150 mm thick ($k=7.32 \text{ kJ/m-hr}^\circ\text{C}$) and on the outside with magnesia bricks 200 mm thick ($k=18.84 \text{ kJ/m-hr}^\circ\text{C}$). The inside and outside surfaces of the wall are at temperature 650°C and 125°C respectively. Make calculations for the heat loss from unit area of the wall.
It is required that the heat loss be reduced to 10 MJ/hr by means of air gap between steel and magnesia bricks. Estimate the necessary width of air gap if the thermal conductivity for air is $0.126 \text{ kJ/m-hr}^\circ\text{C}$.
12. A 30 cm thick furnace wall ($k=1.5 \text{ W/mK}$) has inner surface temperature of 1250°C . If the heat transfer co-efficient on the outer surface is prescribed by the relation $h_o=8+0.09 \Delta t$, where Δt is the temperature difference between the outer wall surface and surroundings, calculate the rate of heat loss per unit area of the wall. Take surrounding temperature as 30°C .
13. A 3 mm thick metal plate having thermal conductivity $k=98.6 \text{ W/mK}$, is exposed to vapor at 100°C on one side and cooling water at 30°C on the opposite side. The heat transfer co-efficient are: $h_i=14200 \text{ W/m}^2\text{K}$ on the vapor side; $h_o=2325 \text{ W/m}^2\text{K}$ on the water side. Determine the rate of heat transfer, the overall heat transfer co-efficient and the drop in temperature at each side of heat transfer.
14. Saturated steam at 110°C flows inside a copper pipe (thermal conductivity 450 W/mK) having an internal diameter of 10 cm and an external diameter of 12 cm. The surface resistance on the steam side is $12000 \text{ W/m}^2\text{K}$ and that on the outside surface of pipe is $18 \text{ W/m}^2\text{K}$. Determine the heat loss from the pipe if it is located in space at 25°C . How this heat loss would be affected if the pipe is lagged with 5 cm thick insulation of thermal conductivity 0.22 W/mK ?
15. An insulated steam pipe of 16 cm diameter is covered with 4 cm layer of insulation ($k=0.9 \text{ W/mK}$) and carries process steam. Determine the percentage change in the rate of heat loss if extra 2 cm thick layer of lagging ($k=1.25 \text{ W/mK}$) is provided. Given that surrounding temperature remains constant and the heat transfer co-efficient for both the configuration is $12 \text{ W/m}^2\text{K}$.
16. A steam main 75 mm inside diameter and 90 mm outside diameter is lagged with two successive layers of insulation. The layer in contact with the pipe is 38 mm asbestos and the asbestos layer is covered with 25 mm thick magnesia insulation. The surface co-efficient for inside and outside surfaces are $227 \text{ W/m}^2\text{K}$ and $6.8 \text{ W/m}^2\text{K}$ respectively. If the steam temperature is 375°C and the ambient temperature is 35°C , calculate the steady state loss of heat from steam for 60 m length of pipe. Also work out the overall heat transfer co-efficient based on the inside and outside surfaces of the lagged steam main. [$k(\text{pipe material})=45 \text{ W/mK}$; $k(\text{Asbestos})=0.14 \text{ W/mK}$; $k(\text{magnesia insulation})=0.07 \text{ W/mK}$].
17. The hot combustion gases of a furnace are separated from the ambient air and its surroundings, which are at 25°C , by a brick wall 0.15 m thick. The brick has a thermal conductivity of 1.5 W/mK and a surface emissivity of 0.8. Under steady state conditions, an outer surface temperature of 100°C is measured. Free convection heat transfer to the air adjacent to the surface is characterized by a convection co-efficient of $h=10 \text{ W/m}^2\text{K}$. What is the inner surface temperature?

18. The surface is covered by a 5 cm thick layer of ice ($k = 2.23 \text{ W/mK}$) when the ambient temperature is -12.5°C . A thermocouple embedded on the upper surface of the ice layer indicates a temperature of -5°C . Assuming steady state conditions in the ice and the liquid submerged at the bottom surface of the ice layer, Find the heat transfer coefficient at the upper surface. Also find out the heat loss per unit area. Subcooling
 32.100°C
19. A surface having an area of 1.5 m^2 and maintained at 300°C exchanges heat by radiation with another surface at 40°C . The value of factor due to the geometric location and emissivity is 0.52. determine:
- Heat lost by radiation.
 - The value of thermal resistance.
 - The value of equivalent convection coefficient.
20. A carbon steel plate (thermal conductivity = $45 \text{ W/m}^\circ\text{C}$) $600 \text{ mm} \times 900 \text{ mm} \times 25 \text{ mm}$ is maintained at 310°C . Air at 15°C blows over the hot plate. If convection heat transfer coefficient is $22 \text{ W/m}^2^\circ\text{C}$ and 250 W is lost from the plate surface temperature by radiation, calculate the inside plate temperature.
21. A steel pipe having an outside diameter of 2 cm is to be covered with two layers of insulation, each having a thickness of 1 cm . The average conductivity of one material is 5 times that of the other. Assuming that inner and outer surface temperatures of the composite insulation are fixed, calculate by what percentage the heat transfer would be reduced when the better insulating material is next to the pipe than it is away from the pipe.
22. A cylindrical rod, 1.5 m long and 2 cm in diameter, is heated electrically and positioned in a vacuum furnace which has interior walls at 800 K temperature. A controlled amount of current is passed through the rod and its surface is maintained at 1000 K . Calculate the power supplied to the heating rod if its surface has an emissivity of 0.9.
23. Two opposed, parallel infinite planes are maintained at 420 K and 480 K respectively. Calculate the net heat flux between these planes if one has an emissivity of 0.8 and other has 0.7. Does it matter which plate has which emissivity. How this heat flux will be effected if
- The temperature difference is doubled by raising 480 K to 540 K
 - The planes are assumed to be black
24. A hot fluid is being conveyed through a long pipe of 4 cm outer diameter and covered with 2 cm thick insulation. It is proposed to reduce the conduction heat loss to the surroundings to one third of the present rate by further covering with same insulation. Calculate the additional thickness of the insulation.

Note: Answers have been provided on page 4

Answer:

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1. 2398.34 kJ/h
2. 58.8 mm
3. 40.66 %
4. 45%
5. 254.67 W/m, 71.56 %
6. 25 mm
7. 2.335 kW
8. 21.78 kW
9. 0.94 W/m²K
10. 486.4 W, 20.3°C
11. 16795 kJ/h, 2.676 mm
12. 5116.2 W
13. 1.31×10^5 , 1882.86, 9.28, 4°C, 56.7°C
14. 578.23 W/m, 164.41 W/m, 71.6%
15. 2.06%
16. 13321.24 W, $U_i=62.19$ W/m²K, $U_o=0.962$ W/m²K
17. 352°C
18. 18.58 W/m²K, 139.3 W/m²
19. 4343.082 W, 0.0599 k/W, 11.136 W/m²K
20. 313.86°C
21. 16.06%
22. 2839.51 W
23. a) heat flux increases by 2.45 times b) heat flux = 1245.52 W/m²
24. 12cm